

Abstract

Multi-agent AI systems are deployed in production, but the measurement science for how their safety failures interact across modes does not yet exist. Our Stage 2b pilot is anchored by a pre-registered self-falsification: the cross-sectional-signal mechanism we initially proposed for LLM-agent Bertrand pricing was reversed by a pre-registered text-based test (Addendum #5, RFC 3161-stamped 2026-05-04 03:38:12 GMT) and the refined per-other-agent attention-density reading then survived a pre-committed talk-volume residualization (Addendum #6, stamped 2026-05-04 05:36:42 GMT) — we apply the same falsification discipline to our own primary mechanism that we apply to the systems we measure, distinguishing this program from concurrent algorithmic-collusion work (Calvano et al., 2020; Cao, 2026). Individual phenomena are well-documented in the literature: failure amplification by $17.2\times$ in decentralized configurations (Kim et al., 2025), 41–86.7% failure rates across production frameworks (Cemri et al., 2025), and constraint-violation rates reaching 71.4% under optimization pressure (Li et al., 2025) — yet zero of 32 multi-agent evaluation papers we surveyed quantify cross-failure-mode propagation.

We propose a 36-month empirical program mapping directly to Aim 3 and Aim 3.2 Question 35 — when and under what conditions do AI agents learn to collude, and what are the implications for oversight. Three pre-registered hypotheses unify under a cascade model: failure amplification creates network pressure that enables emergent collusion (H1 to H2), collusion accelerates norm erosion by demonstrating violations go undetected (H2 to H3), and erosion removes constraints that would dampen amplification (H3 to H1). Experiments span five topologies, 2 to 256 agents, nine model families (five open-weight, four frontier APIs), and three communication protocols. We measure topology-specific Failure Amplification Factors with bootstrap CIs, collusion detection latency via partial information decomposition synergy, norm-erosion thresholds via changepoint detection, and cross-failure-mode interaction coefficients via cross-classified Bayesian SEM with Rosenbaum sensitivity bounds. A $2\times 2\times 2$ knock-out factorial isolates directional causation. Our Stage 2b pilot also produces a regime-disjoint Stigler reframe — distinct from Stigler’s price-theoretic free-rider mechanism because LLM agents are a different agent class with different ontologies (preprint §5.2 verbatim) — applied to LLM-agent Bertrand pricing, with bimodal $k = 2$ basin structure recovered at $n = 5$ (7/10 gap-statistic resamples; 30/30 basin-stable) and not at $n = 2$ within the $n_{\text{rep}} = 15$ envelope. This basin-asymmetry reading is **EXPLORATORY** pending Stage 3 — small-sample by design — and is converted into a Stage-3-pre-committed prediction via SR-M-8 (monotonic strict-convergence rate across $n \in \{2, 3, 4, 5\}$ with intermediate counts Wilson-bracketed by the endpoints; RFC-3161-protected through the Addendum #6 stamp chain). SR-M-6 and SR-M-7 remain **EXPLORATORY** pending Stage 3 SciPy verifier port per Addendum #5/#6 §C. Deliverables include the open-source ColludeBench-Cascade open-source platform, 8–12 peer-reviewed publications, 10,000+ annotated traces, and policy translation aligned with EU AI Act Article 73 (binding August 2, 2026). H.H.A. Applied Research Institute is a California Public Benefit Nonprofit Corporation granted 501(c)(3) status by the IRS (EIN 41-4991887), model-provider-independent, with 90.9% of the \$4,356,500 cash envelope directed to direct research (compute capacity requested in-kind via Schmidt’s “Access to Resources” channel). # The Problem and the Measurement Gap

Multi-Agent Architectures Are Already in Production

The shift to multi-agent AI is complete at every major platform vendor. Microsoft Copilot Studio, Google Vertex AI Agent Builder, Amazon Bedrock multi-agent collaboration, and Salesforce Agentforce all ship production multi-agent orchestration making consequential decisions in pricing, compliance, hiring, medical triage, and financial risk (Microsoft Build 2025; Google Cloud 2025; AWS re:Invent 2024; Salesforce 2024). The safety science governing these systems was built for a different architecture. NIST AI RMF, EU AI Act conformity assessments, and Executive Order 14110 red-teaming each assume one system, one output, one accountability chain. They evaluate components, not the network that emerges when components interact — and that is precisely where multi-agent deployment is consequential.

Three Findings, One Threat Surface

Failure amplification is measurable, architecture-dependent, and extreme. Kim et al. (2025; arXiv:2512.08296) evaluated 180 configurations across five agent architectures, four benchmarks, and three LLM families, reporting $17.2\times$ amplification in independent multi-agent configurations versus $4.4\times$ with centralized coordination. A 1% individual-agent failure becomes 17.2% in a decentralized network — propagation dynamics that did not exist at the component level.

Failure rates in deployed frameworks are catastrophic. Cemri et al. (2025; arXiv:2503.13657) constructed MAST from 1,642 traces across AutoGen, CrewAI, LangGraph, and four other production frameworks; failure rates ranged from 41% to 86.7% across 14 distinct failure modes (Cohen’s $\kappa = 0.88$). These are normal operating conditions, not adversarial stress tests.

Capable models violate constraints knowingly. Li et al. (2025; arXiv:2512.20798) report 1.3–71.4% violation rates across twelve frontier models on ODCV-Bench’s 40 scenarios, with deliberative misalignment: agents violating in

operational contexts correctly identify the same actions as unethical in retrospective probes. This is structural under optimization pressure that multi-agent networks intensify through competition, delegation, and peer observation.

These findings are mutually reinforcing. Amplification creates network pressure (Reid et al., 2025; arXiv:2508.05687, “a collection of safe agents does not guarantee a safe collection of agents”); pressure drives erosion (Rath, 2026; arXiv:2601.04170, median drift onset at 73 interactions, 53.2% susceptibility by 500 interactions); erosion removes the constraints that would otherwise contain emergent collusion (Bracale Syrnikov et al., 2026; arXiv:2601.11369, governance graphs reduce collusion from tier 3.10 to 1.82, $d = 1.28$).

The collusion mechanism extends a long industrial-organization literature on algorithmic supra-competitive pricing: Q-learning duopolies converge to multimodal terminal-state distributions in repeated Bertrand and Cournot games (Calvano et al., 2020; Waltman & Kaymak, 2008; Wunder, Littman & Babes, 2010); algorithmic pricing is empirically a regulator-relevant phenomenon (Asker, Fershtman & Pakes, 2021; Assad et al., 2024); and existing competition law typically requires evidence of explicit agreement that self-learning agents do not produce (Ezrachi & Stucke, 2016). LLM agents introduce a third algorithmic family that carries no shared reward signal, no explicit value function, no online weight updates, and no exploration parameter — whether the basin-of-attraction structure characterizing Q-learner outcomes extends to LLMs is largely uncharacterized in the literature. Concurrent benchmark work — most directly the multi-domain steganographic-collusion benchmark of Tailor (2025; arXiv:2510.04303) and the DCOP-based collusion audit of Nakamura et al. (2026; arXiv:2602.15198) — addresses complementary scope; our basin-of-attraction characterization at fixed Bertrand pricing (preprint at github.com/HHA-Applied-Research-Institute/colludebench-cascade) is complementary, not duplicative.

The Measurement Gap

We surveyed 32 multi-agent evaluation papers and 13 funding sources. Twenty-six measure miscoordination. Five measure collusion. **Zero measure the interaction between failure modes.** A representative slice across MACHI-AVELLI, SafeAgentBench, AgentHarm, R-Judge, MAST, ColludeBench-v0, Colosseum, ODCV-Bench, MAEBE, MultiAgentBench/MARBLE, MultiAgentFraudBench, AgentMisalignment, Secret Collusion/CASE, the PID Emergence framework, and Governance Graphs — all evaluate component failure modes; none measure cross-mode propagation coefficients (Hammond et al., 2025, arXiv:2502.14143, document the same gap in the Cooperative AI Foundation taxonomy). The component science is strong. The integration science does not exist.

The funding landscape mirrors this gap. Open Philanthropy committed approximately \$50M to technical AI safety in 2024; DARPA requested \$22M for ASIMOV; Schmidt Sciences funded 27 safety projects through its 2025 AI Safety Science Program. None of these investments support research on how collusion, amplification, and norm erosion interact when they co-occur in the same network.

The Policy Vacuum

The regulatory frameworks governing AI safety are structurally unprepared for multi-agent failure. EU AI Act Article 73, binding August 2, 2026, “assumes only one system will contribute to a high-risk incident” (TechPolicy.Press, 2025); the amendment process requires empirical evidence of multi-agent failure patterns that does not yet exist in regulator-operationalizable form. NIST AI RMF recognizes inter-agent dependencies as a risk factor but provides no multi-agent testing methodology. Executive Order 14110’s red-teaming mandate evaluates individual models in isolation. The 2025 RAND/Schmidt/Palo Alto Networks report on agent-ecosystem security (Nichols, Nevo, Greaves) and the 2026 International AI Safety Report (Bengio et al., 30+ countries) both name the gap and both lack the measurement infrastructure to close it. # Research Questions and Hypotheses

Overarching question. How do safety failures propagate, amplify, and compound through networks of cooperating AI agents — and can we detect emergent collusion, goal drift, and norm erosion before they become irreversible? Network-level failures are qualitatively distinct from their single-agent constituents (Kim et al., 2025; Cemri et al., 2025). We test whether they interact and whether that interaction produces cascade dynamics worse than the sum of their parts.

Construct definitions (each grounded in a validated instrument). *Safety failure*: MAST 14-mode taxonomy at $\kappa = 0.88$ (Cemri et al., 2025) with severity scaled benign / propagated / amplified / cascading. *Collusion*: Colosseum DCOP-based regret (Nakamura et al., 2026) plus PID synergy (Riedl, 2025; positive synergy implies emergent coordination not attributable to individual behavior). *Norm erosion*: statistically significant negative trend in the Constraint Compliance Rate with PELT changepoint onset (Killick et al., 2012) and peer-influence coefficient via Granger temporal precedence; mandated versus incentivized violations distinguished by ODCV-Bench’s dual probe (Li et al., 2025).

H1 — Failure Amplification

Safety failures in individual agents amplify in multi-agent networks, with magnitude dependent on topology: $\text{FAF}(\tau, n) = \bar{f}^N / \bar{f} > 1$ for all tested topologies. **Success criterion:** statistically significant $\text{FAF} > 1.0$ (one-sided, p

< 0.01 , Bonferroni-corrected across five topologies), 95% bootstrap CIs; cascade-size distribution exponent $\tau < 2$ via Clauset-Shalizi-Newman MLE. **Anchor:** Kim et al. (2025) report FAF = 17.2 (independent) and 4.4 (centralized) across 180 configurations; Xie et al. (2026) demonstrate single-seed cascades; Huang et al. (2024) report 5.5% degradation hierarchical versus 23.7% flat — confirming topology dependence.

H2 — Collusion Detection

Emergent collusion in cooperative networks produces detectable behavioral signatures — positive PID synergy and anomalous mutual information profiles — before explicit coordination occurs: $t_{\text{syn}} < t_{\text{coord}}$. **Success criterion:** AUC > 0.85 across ColludeBench-v0’s three domains (pricing, auction, peer review) under cross-domain test, with FPR $< 10^{-3}$ as the locked target and FPR $< 10^{-2}$ as the Stage 3 success-criterion floor (validated against 10,000 honest-play runs per Tailor, 2025); cross-domain generalization measured at fixed threshold without re-tuning. **Aim 3.2 Q35 mapping:** H2 is the direct empirical answer to “when and under what conditions do AI agents learn to collude, and what are the implications for oversight.” Stage 2b of our pilot has produced one component of this evidence — a regime-disjoint Stigler reframe (in the sense of Calvano et al. 2020 algorithmic-collusion-as-equilibrium-selection) applied to LLM-agent Bertrand-pricing dynamics with bimodal $k = 2$ basin structure recovered at $n = 5$ (7/10 gap-statistic resamples; 30/30 basin-stable) and not at $n = 2$ within the $n_{\text{rep}} = 15$ envelope (preprint at github.com/HHA-Applied-Research-Institute/colludebench-cascade). The presence/absence contrast at $n_{\text{rep}} = 15$ is **EXPLORATORY** by design (Wilson 95% CI on 7/10 is wide); SR-M-8 converts it into a Stage-3-pre-committed prediction — monotonic strict-convergence rate across $n \in \{2, 3, 4, 5\}$ with intermediate counts Wilson-bracketed by the endpoints, RFC-3161-protected through the Addendum #6 stamp chain — so a non-monotonic or non-bracketing Stage 3 curve rejects the framework. The cross-sectional-signal mechanism we initially proposed was empirically reversed by a pre-registered text-based test, and the refined reading survived talk-volume residualization (SR-M-6 and SR-M-7 are **EXPLORATORY** status — primary TypeScript analyzer fired the pre-committed REVERSAL and REVERSAL-SURVIVES-RESIDUALIZATION branches; SciPy reference-verifier promotion to PASS is deferred to Stage 3 per Addendum #5/#6 §C). We apply the same falsification discipline to our own primary mechanism that we apply to the systems we measure (Asker, Fershtman & Pakes, 2021; Ezrachi & Stucke, 2016).

H3 — Norm Erosion

Safety constraints degrade over extended multi-agent interaction through normalization of deviance, with erosion rate determined by peer-influence dynamics and an identifiable threshold beyond which recovery is infeasible without external intervention: $d\text{CCR}_i/dt < 0$ with changepoint t^* ; peer influence $\rho > 0$; post-changepoint behavior exhibits deliberative misalignment per ODCV-Bench dual-probe. **Success criterion:** identifiable threshold with $>90\%$ reproducibility across model families; ρ at $p < 0.01$; clear empirical distinction between mandated and incentivized violation rates across ODCV-Bench’s 40 scenarios. **Anchor:** Rath (2026) reports median drift onset at 73 interactions; Li et al. (2025) report 1.3-71.4% violation rates across 12 frontier models; Bracale Syrnikov et al. (2026) demonstrate prompt-only constraints fail under optimization pressure — Hadfield-Menell inverse-reward-design framing.

Cascade Hypothesis (Primary Contribution)

H1, H2, H3 form a directed causal cascade: H1 creates pressure under which H2 becomes individually rational; H2 accelerates H3 by demonstrating violations go undetected; H3 feeds back into H1 by removing constraint-adherent behavior that would dampen propagation. We specify the cascade as a structural causal model (Pearl, 2009):

$$\tau \rightarrow \text{FAF} \rightarrow \text{Syn} \rightarrow \Delta\text{CCR}$$

The DAG encodes two testable conditional independences: $\Delta\text{CCR} \perp\!\!\!\perp \tau \mid \text{FAF}, \text{Syn}$ (full mediation via amplification and collusion) and $\text{Syn} \perp\!\!\!\perp \tau \mid \text{FAF}$ (topology drives collusion only through amplification pressure). Violations indicate direct paths bypassing the cascade. **Success criterion:** bootstrapped indirect-effect CI excludes zero (10,000 resamples); cascade model fits significantly better than parallel-effects and reverse-cascade alternatives via WAIC; Rosenbaum sensitivity at $\Gamma \geq 2.0$ for each link. The 2x2x2 knock-out factorial (§5) provides experimental evidence for directionality.

Null-result value. Each hypothesis has defined null value. H1 null ($\text{FAF} \leq 1.0$) shows networks naturally contain failures, contradicting the prevailing assumption. H2 null implies collusion emerges discontinuously, requiring different detection architectures. H3 null endorses individual-agent alignment as sufficient. Cascade null implies modular interventions, simplifying safe multi-agent engineering. Pre-registration on OSF Registries with RFC 3161 trusted timestamps locks hypotheses, endpoints, pipeline, and stopping rules before any data collection — making null results publishable. # Methodology and Experimental Design

Our experimental design rests on three contributions. (1) The first **causally-identified** cross-failure-mode interaction estimation in multi-agent networks under Heng-Kang generalized sensitivity bounds — going beyond co-occurrence frequencies (e.g., the 14-mode MAST taxonomy of Cemri et al., 2025) by running amplification, collusion, and erosion measurement pipelines simultaneously in the same agent networks with explicit causal identification. (2) An integrity

scaffold — RFC 3161-stamped pre-registration, cross-toolchain reference verifier, named-expert adversarial Council review — described here and in §12 as the team’s *internal methodology*, not as authorship. (3) A 2x2x2 knock-out factorial that adjudicates cascade directionality experimentally.

5.1 Factorial Design

The primary factorial spans five network topologies (peer-to-peer mesh, hierarchical tree, hub-and-spoke star, ring, task-adapted DAG via AgentNet, arXiv:2504.00587), five agent scales (2, 8, 32, 128, 256), nine model families (five open-weight: Llama 4 Scout, GPT-OSS-120B, Qwen3-235B, DeepSeek V3.2, Mistral Large 2; four frontier APIs: Claude Opus 4.6, GPT-5.5, Gemini 3 Pro, Claude Sonnet 4.6 — versions current at submission, with successor models adopted at experiment-execution time as released), and three communication protocols (open broadcast, private channels, monitored channels with independent oversight per Fan et al., 2025, arXiv:2505.11642). Open-weight models serve as model organisms — fully instrumentable with attention patterns, hidden states, safety-head activations, and expert routing visible — providing the mechanistic understanding that generalizes to the closed-weight frontier APIs. Mistral Large 2 is the dense, non-MoE control. Full factorial at 2, 8, 32 agents; Resolution IV fractional factorial at 128 and 256 (Groemping, 2014) targeting main effects and cascade knock-out cells. Six validated instruments anchor measurement: MAEBE (Erisken et al., 2025; arXiv:2506.03053), ColludeBench-v0 (Tailor, 2025; arXiv:2510.04303), Colosseum (Nakamura et al., 2026; arXiv:2602.15198), MAST (Cemri et al., 2025), ODCV-Bench (Li et al., 2025), and the PID Emergence framework (Riedl, 2025). Our open-source distribution **ColludeBench-Cascade** (D1) extends Tailor’s ColludeBench-v0 with cross-failure-mode cascade measurement across all six.

Every cell instruments the same network for all three failure modes concurrently, recording at 50-turn intervals: FAF via paired comparison against individual baselines, PID synergy scores via time-delayed mutual information decomposition, and CCR trajectory via ODCV-Bench’s dual-probe. Causal identification rests on three convergent strategies: (i) temporal precedence via vector autoregression on the (FAF, Syn, Δ CCR) time series, (ii) experimental manipulation via the 2x2x2 knock-out factorial (§5.4), and (iii) Rosenbaum sensitivity bounds with the Heng–Kang generalized framework for non-binary treatments (arXiv:2403.14152) and the Dalal–Tchetgen Tchetgen partial-identification extension (arXiv:2508.13946).

5.2 Integrity Scaffold (Contribution 2)

The integrity scaffold is the team’s *internal methodology* for maintaining analytic-claim integrity before, during, and after any analyzer runs. Four layers, hierarchical not symmetric:

RFC 3161 trusted-timestamp chain. All pre-registration documents and addenda are stamped before data analysis begins. Stage 2b ran with eight FreeTSA stamps (two pre-registrations, six addenda), all **Status: Granted**, each topologically anterior to the analyzer-output commit it governs (stamp time strictly precedes git-commit-time). The cryptographic lock is the strongest integrity primitive available for analysis-plan locking; the verification and adversarial layers reinforce it but cannot substitute for it. We replicate this scaffold for every Stage 3 and Stage 4 analyzer in this proposal.

Pre-committed Stage Reasoning Mechanistic (SR-M) registry. Each load-bearing claim has a registry entry with explicit falsification conditions and pre-committed rescope branches stamped before the analyzer runs. Stage 2b shipped with eight SR-M entries; one (SR-M-6) recorded a self-falsification — the cross-sectional-signal mechanism we initially proposed was reversed by a pre-registered text-based test (Addendum #5 stamped 2026-05-04 03:38:12 GMT, 25 seconds before analyzer output). A second (SR-M-7) recorded that the reversal survived talk-volume residualization (Addendum #6, stamp 2026-05-04 05:36:42 GMT). The §5.4 Clause 4 Stage 3 monotonicity test (SR-M-8) is RFC-3161-protected through the Addendum #6 stamp chain.

Cross-toolchain reference verifier. The TypeScript primary analyzer is independently re-implemented in Python with SciPy reference statistical primitives. The verifier runs against the same canonical dataset and returns **ALL CLAIMS REPRODUCE: True** across the six pre-registered claim categories where the SciPy verifier has shipped (SR-M-1 through SR-M-5 plus the cross-toolchain reproduction claim itself, all PASS) at $|\Delta| < 5 \times 10^{-3}$; SR-M-6 and SR-M-7 are EXPLORATORY status (primary TypeScript analyzer fired the pre-committed REVERSAL and REVERSAL-SURVIVES-RESIDUALIZATION branches; SciPy verifier promotion to PASS is deferred to Stage 3 per Addendum #5/#6 §C). SR-M-8 is PRE-REGISTERED for Stage 3 falsification (rate-vs-n monotonicity). This is procedural reproduction, not mere replication: a Fisher’s-exact denominator bug found pre-submission via this procedure would have invalidated a primary claim.

Named-expert adversarial Council. Five iterations against persona panels (DataScientist + Statistician/Algorithmist, CSProfessor, Physicist, IO Domain Expert, Devil’s Advocate, Methods/Reproducibility, Consistency, Writing Quality). Stage 2b’s Council exited CONDITIONAL after iter-5 with documented user-override rationale; ~32 substantive findings (CRITICAL + HIGH) were applied across iterations. The full click-to-verify map for every empirical claim — code path, data path, RFC 3161 stamp, Council certificate state — is published at `verification/claims-map.md` in the public GitHub repository.

This scaffold is described as the team’s *practice*, not as authorship. The 2025 RAND/Schmidt/Palo Alto Networks report (Nichols, Nevo, Greaves) names “manipulation of activity records” as a first-order agent-ecosystem risk; integrity scaffolds of this stringency are precisely the response.

5.3 Knock-Out 2x2x2 Factorial

The cascade hypothesis requires experimental evidence for directionality, not correlation. We cross presence/suppression of each failure mode across eight cells. **Suppression mechanisms:** H1 — hierarchical topology with redundancy plus circuit breakers (DeepMind’s 4.4x containment); H2 — governance-graph enforcement (reduces collusion from tier 3.10 to 1.82, $d = 1.28$; Bracale Syrnikov et al., 2026) plus selective paraphrase filters; H3 — agent context reset every 100 turns with task state preserved through an external memory store agents cannot modify. **Critical contrasts:** (a) H1_suppressed main effect on H2 metrics — does suppressing amplification reduce collusion emergence? (b) H2_suppressed main effect on H3 metrics — does suppressing collusion slow erosion? (c) H1:H2 interaction on H3 — does suppressing both upstream modes reduce erosion *more than* the sum of individual suppressions, providing superadditive cascade evidence? Sample size: 8 cells $\times$ $n = 30$ runs = 240 runs on the maximum-cross-mode-variance topology-protocol cell from Class A. The §5.4 falsification clauses preregistered in our preprint commit the cascade to a Stage 3 monotonicity test: basin-width-by-agent-count predicts a monotonic strict-convergence rate across $n \in \{2, 3, 4, 5\}$ with intermediate counts Wilson-bracketed by the endpoints; non-monotonic or non-bracketing curves reject the framework (SR-M-8).

5.4 Stage 3 Instrumentation Extension

Stage 3 extends the Stage 2b pipeline by extending Addendum #1 §F’s two stamped trace-schema fields (`model_version`, `region`) with five Stage-3-pre-committed additions (`ttft_ms`, `tpot_ms`, `cache_hit_tokens`, `cache_miss_tokens`, `auth_handshake_ms`) — the seven-field schema is finalized at Stage 3 pre-registration (RFC 3161 stamped before any new data collection). The schema isolates LLM-serving variability (Kwon et al., 2023, paged attention; Zheng et al., 2023, SGLang; Holmes et al., 2024, DeepSpeed-FastGen) from reasoning-depth variability (Snell et al., 2024, test-time compute) — a confound Stage 2b explicitly does not adjudicate, and the seven-field schema is the pre-committed Stage-3 confirmatory test for it; the Stage-2b latency-reasoning-depth correlation is therefore **EXPLORATORY** pending this deconfound. Full Stage-2b experimental detail is in the preprint at github.com/HHA-Applied-Research-Institute/colludebench-cascade.

5.5 Power, Pre-Registration, and Software

Simulation-based power via `simr` (Green & MacLeod, 2016) accounts for clustering inherent in network experiments; ICC = 0.15 primary, 0.05 and 0.30 sensitivity. With $DEFF = 1 + (m-1) \cdot ICC$, allocations are: $n = 50$ runs/cell at 2-32 agents (power 0.82-0.94, $d = 0.5$, $\alpha = 0.005$); $n = 100$ for H2-critical cells where the smallest effects reside (MAEBE peer pressure $d = 0.3$); fractional factorials at 128 and 256 agents target main effects and knock-out conditions. Hierarchical gatekeeping preserves family-wise error: H1 gateway at $p < 0.005$, then H2, then H3, then cascade; Holm-Bonferroni within each. Two pre-planned interim analyses at $n = 40$ and $n = 80$ use Lan-DeMets O’Brien-Fleming spending (cumulative α 0.00005, 0.0030, 0.0100; computed via `rpact`); binding futility at conditional power < 0.20 . The full analysis pipeline — utterance-level PID, window-level cross-classified Bayesian SEM in `brms/blavaan`, session-level PELT and BOCPD changepoint detection — is jointly estimated in Stan, propagating uncertainty across levels. Pre-registration on OSF Registries locks: hypotheses, primary endpoints, full pipeline, knock-out contrasts, stopping rules, $\Gamma = 2.0$ credibility threshold, exclusion criteria, exact software versions, blinding protocol, and prior-sensitivity specifications (weakly informative primary, informative pilot-based, diffuse). All elements are RFC 3161-stamped before data collection. # Expected Outcomes and Deliverables

Six named deliverables, each with timeline, target venue, success criterion, and null-result value. Schmidt requires proposals “state clearly what would happen if the project succeeds, and what we learn if it fails.” We answer both for every deliverable.

D1 — ColludeBench-Cascade open-source platform. Configurable populations (2–256 agents), five topologies, three communication protocols, built-in measurement pipelines for all three failure modes, model-agnostic interfaces, deterministic replay, MAST-compatible trace export, real-time cascade visualization. *Timeline:* alpha M18, v1.0 M30. *Venue:* GitHub release + NeurIPS 2028 D&B systems paper. *Success:* concurrent 256-agent experiments across two+ model families; ≥ 3 external groups adopt within 12 months. *Null value:* serves as internal infrastructure producing all other deliverables regardless of external adoption.

D2 — Topology-specific failure amplification map. Quantitative FAF estimates for five topologies (centralized hub-spoke, mesh, hierarchical tree, ring, DAG), each with 95% CIs, p-values for topology effects, and propagation-pathway diagrams. *Timeline:* preliminary M12, complete M24. *Venue:* ICML 2027 / ICLR 2028. *Success:* statistically significant amplification ($p < 0.01$, Bonferroni) across all five topologies; calibrated against DeepMind baselines ($4.4\times/17.2\times$); predictive validity via leave-one-topology-out cross-validation. *Null value:* topology-independent FAF

redirects intervention strategy from architectural design toward agent-level containment — itself consequential.

D3 — Validated collusion detection system (Aim 3.2 Q35 direct). Real-time monitoring pipeline integrating cross-run mutual information estimation, permutation invariance testing, watermark variance, and fairness-aware acceptance bias detection into a unified meta-test. *Timeline:* prototype M12, validated M18. *Venue:* NeurIPS 2027 / ACL 2028. *Success (per §4 H2):* AUC > 0.85 across ColludeBench-v0’s three domains (pricing, auction, peer review) under cross-domain test, with FPR < 10^{-3} as the locked target and FPR < 10^{-2} as the Stage 3 success-criterion floor; cross-domain generalization measured at fixed threshold without re-tuning; channel-capacity (bits/turn) estimates for covert coordination. *Null value:* establishes empirical bound on early-warning capability — critical information for deployers needing to understand monitoring-based safety limits.

D4 — Norm erosion thresholds for 5+ model families. Quantitative thresholds at which constraint adherence degrades past recovery, per model family, with curve shape (linear, exponential, S-curve, step), inflection points via PELT/BOCPD, and peer-influence coefficients. *Timeline:* 2 open-weight families M15; 5+ (incl. frontier) M24. *Venue:* ICLR 2028 / AIES 2028. *Success:* identifiable inflection point with >90% reproducibility; peer-influence $p < 0.01$ for ≥ 3 families; mandated-vs-incentivized distinction per ODCV-Bench dual-probe. *Null value:* highly stochastic erosion indicates universal operational scheduling rules are infeasible — itself decision-relevant.

D5 — Cross-failure-mode interaction coefficients (PRIMARY DELIVERABLE). First **causally-identified** measurement of how amplification, collusion, and norm erosion compound. Reported as interaction terms in cross-classified Bayesian SEM: does H1 amplification predict H3 erosion rate? Does the cascade compound exceed the sum of individual effects? *Timeline:* preliminary M20, validated M30. *Venue:* NeurIPS 2028; *Science / Nature Machine Intelligence* if cascade confirmation is robust. *Success:* bootstrapped indirect-effect CI excludes zero (10,000 resamples) for ≥ 2 of three pairwise pathways under §5.5’s hierarchical gatekeeping (final cumulative $\alpha = 0.0100$ per Lan–DeMets O’Brien–Fleming spending; Holm–Bonferroni within stage); cascade model fits significantly better than parallel-effects and reverse-cascade alternatives via WAIC model comparison; compound cascade exceeding sum-of-individuals by measurable margin; replication across ≥ 2 model families; Rosenbaum sensitivity at $\Gamma \geq 2.0$ per cascade link; predictive validity — given measured amplification and collusion, can we forecast erosion before it manifests? *Power:* simulation-based power calculation pre-registered alongside the §5.5 hierarchical-gatekeeping spec — at the §11 H2-critical-cell allocation under primary ICC = 0.15, achieves ≥ 0.80 power to detect a medium interaction effect ($d_{\text{int}} = 0.35$) at the §5.5 cascade-stage α -allocation (final cumulative $\alpha = 0.0100$), with explicit sensitivity at ICC $\in \{0.05, 0.30\}$. Where the DEFF-corrected power for $d_{\text{int}} = 0.30$ falls below 0.80, the cell is escalated to fractional-factorial coverage at 128/256 agents with re-allocated runs (§5.5). *Null value:* non-significant interaction coefficients are strong evidence that individual-mode interventions suffice — among the most consequential findings possible in multi-agent safety, published with equal rigor.

D6 — Field-shaping research outputs. Three additional outputs reach the broader research, policy, and infrastructure communities. (a) 8–12 peer-reviewed publications across NeurIPS, ICLR, ICML, ACL 2027–2029 + survey paper at *Nature Machine Intelligence* + ColludeBench-Cascade systems paper at NeurIPS 2028 D&B; success: ≥ 8 papers at top-tier venues by M36, ≥ 2 with findings validated across ≥ 3 model families. (b) *Policy translation:* 2–3 briefs mapping topology-specific findings to EU AI Act Article 73 (binding 2026-08-02), one technical guidance document aligned with NIST AI RMF, intent to engage NIST Dioptra on cascade-detection methodology; success: ≥ 1 brief cited in regulatory consultation by M36. (c) *Open dataset:* 10,000+ MAST-annotated multi-agent interaction traces with reproducibility metadata, MAST-compatible JSONL, NeurIPS 2028 D&B + Hugging Face Datasets; success: kappa > 0.80 for failure-mode classification, ≥ 2 external research groups adopt within 12 months. *Null value:* even with non-confirming hypotheses, the corpus is the largest publicly available multi-agent safety interaction dataset, and the policy briefs bridge technical research to governance independent of our framework. # Why \$5M-Tier — The Scale Justification

Cascade dynamics are scale-dependent by definition: they emerge from network effects that do not exist at small agent counts. The cash budget is \$4,356,500; compute capacity (8×H100 dedicated + burst, cash-equivalent ~\$556K) is requested via Schmidt’s “Access to Resources” in-kind channel, putting total program resources at the \$5M Tier-2 scale. The table below distinguishes what is achievable at two funding levels — a statement of experimental physics, not a negotiating position.

Dimension	\$1M Cash Budget	\$5M-Tier Program Resources (\$4.36M cash + Schmidt in-kind compute)	Scientific Justification
Agent networks	2-32 agents	2-256 agents	Cascade dynamics exhibit phase transitions at scale; 32-agent networks cannot produce the network-level contagion that characterizes deployments of hundreds of agents
Model families	2 open-weight	5 open-weight + 4 frontier APIs	Architecture-specific vs. architecture-independent failure modes can only be identified through cross-family validation
Duration	24 months	36 months	Longitudinal erosion requires extended observation; 36 months permits three experimental cycles, the minimum for reproducible erosion-curve estimation with inflection-point identification
Compute	4xH100	8xH100 dedicated + burst	256-agent experiments with multiple MoE models require concurrent inference at scale; 4 GPUs make large-network experiments computationally infeasible at the turn counts erosion studies require
Team	5 people	12 people (3 PIs at 25-30%, 3 postdocs, 2 senior engineers, 2 RAs, 0.5 PM)	Three parallel experimental tracks require three postdoc-led teams; 256-agent platform engineering requires two dedicated senior engineers
Knock-out design	Not possible	2x2x2 factorial with pathway-blocking interventions	Causal identification of cascade mechanisms requires interventions that suppress one failure pathway while leaving others intact; cost-prohibitive below \$4M
Publications	4-6 papers	8-12 papers across NeurIPS, ICML, ICLR, AIES	Three parallel tracks produce three independent publication streams; cross-track integration adds 2-3 contributions
Policy translation	None	EU AI Act + NIST translation via governance subaward	Tier 2 expects field-shaping deliverables beyond publications; regulatory translation requires governance expertise the core team does not possess
Open-source platform	Basic dataset release	Full ColludeBench-Cascade benchmark with 3-year maintenance	The ColludeBench-Cascade platform becomes field infrastructure — standardized environments, configurable populations, built-in metrics — enabling community replication

These differences are qualitative, not incremental.

Where the \$5M goes

Personnel — \$3,035,000 (69.7%). Three postdocs (Collusion/Game Theory 36mo; Norm Erosion/Longitudinal 36mo; Detection Methods 24mo from Year 2) execute the three parallel experimental tracks. Two senior engineers (\$1.1M combined) build inter-agent interception middleware, configurable topology controllers, real-time PID measurement pipelines, and 256-agent experiment controllers — infrastructure that does not exist in any current framework. The remaining personnel (3 PIs at 25-30%, 2 RAs, 0.5 PM) provide scientific direction, daily execution, and multi-institution coordination. At \$1.67M/year for ~10.5 FTE the per-researcher cost is below NSF AI Institute norms.

Compute (cash) — \$50,000 (1.1%); 8×H100 cluster + burst requested in-kind via Schmidt’s “Access to Resources” channel (~\$556K commercial-cloud equivalent). Cash covers storage, data-egress, and orchestration not bundled into in-kind GPU allocations. The 8×H100 80GB cluster (640 GB VRAM, 128+ concurrent agent instances at 7B-17B active MoE, INT4/MXFP4) plus burst capacity enables 256-agent peak experiments: at 32 agents one H100 serves all at sub-second latency; at 256 agents over 5,000-turn erosion studies, 4-GPU runtime exceeds 8 hours — infeasible for 8,250+ runs. Post-award CAIS supplementary-access path and cash-reactivation pathway in

compute-strategy.md.

Subawards — \$350,000 (8.0%). Two branches selected via open RFP within the first six months of award; selection criteria published as part of the institute’s open methodology. **Branch X — Game Theory (\$225K):** one postdoc + 0.5 faculty grounding behavioral detection in formal collusion models (Nash equilibria, mechanism design); selection criteria require prior publication on multi-agent strategic interaction. **Branch Y — Governance (\$125K):** 0.25 FTE policy researcher producing NIST AI Safety standards engagement + EU AI Act Article 73 technical-annex contributions + 2–3 policy briefs; selection criteria require documented record in AI governance / policy translation. Schmidt’s 10% indirect-cost cap is built into both selection criteria; selection-after-award matches partner choice to experimental needs as they crystallize from Stage 2b → Stage 3.

Other lines (API credits, travel, ColludeBench-Cascade platform external work, equipment, publications, external methodology consultations) total \$525,500 and are itemized in the attached BUDGET.xlsx.

\$3M Fallback

At \$3M the program retains the cluster but loses burst capacity, retains 2 postdocs but drops the detection-methods track, retains one engineer (forcing sequential platform development), reduces frontier validation from 5 families to 3, shortens longitudinal erosion studies from 5,000 to 2,000 turns, drops policy translation, and releases datasets without the full ColludeBench-Cascade platform. The most consequential loss is the knock-out factorial — the only design that distinguishes genuine cascade effects from the simpler explanation that all three failure modes increase under network-complexity confounding. At \$3M we document that cascade failures exist; we cannot establish causally and mechanistically how they propagate, when they become irreversible, and what practitioners can do to stop them. The scale matches the phenomenon. # Team Qualifications

Institutional Independence

H.H.A. Applied Research Institute is a California Public Benefit Nonprofit Corporation, granted 501(c)(3) tax-exempt status by the IRS (EIN 41-4991887). This is the PIs’ primary research focus; H.H.A. runs no competing program of comparable scope. H.H.A. has no model-provider affiliation, no commercial entanglement, and no conflict of interest with any system under evaluation. Stanford HAI receives Google and Microsoft funding; MIT CSAIL partners with both; Berkeley AI Research operates within Meta’s ecosystem. These relationships create structural conflicts when findings — for example, evidence that a funder’s model exhibits emergent collusion — carry commercial consequences. For trustworthy AI evaluation, evaluator independence is a prerequisite for construct validity, and H.H.A.’s actual sub-10% overhead means approximately \$3.96M of every \$4.36M cash dollar reaches direct research (90.9%) — roughly 1.5x the effective rate of an equivalent university proposal under Schmidt’s 10% indirect cap.

Lead PI: Haedar Hadi

Haedar Hadi directs the overall research program with primary responsibility for measurement-platform architecture, evaluation methodology, and institutional leadership. Hadi brings research-infrastructure experience from Stanford University and the Hoover Institution, managing operations and infrastructure budgets in the \$600,000–\$800,000/year range, coordinating multi-stakeholder programs with fixed-timeline deliverables. Hadi is the architect of the ColludeBench-Cascade measurement platform at the center of this proposal and led the 13-benchmark survey that identified the cross-failure-mode measurement gap motivating this research. Under this proposal Hadi oversees research operations, manages the engineering team, coordinates institutional partnerships, and serves as primary liaison with Schmidt Sciences.

Co-PI: Hass Dhia

Primary role: Experimental design lead; collusion detection methodology (direct Aim 3.2 Q35 alignment — *when and under what conditions do AI agents learn to collude, and what are the implications for oversight*); statistical analysis and power calculations.

Alhassan (Hass) Dhia holds an MS in Biomedical Sciences from Wayne State University School of Medicine; graduate training in anatomy, physiology, and pharmacokinetics established the quantitative-modeling foundation underpinning this proposal’s experimental design. Dhia is the architect of H.H.A.’s research-automation infrastructure, enabling a three-person team to produce literature synthesis, experimental design, and adversarial review at the velocity of a 30-person lab. Dhia leads experimental design and statistical analysis across all three hypothesis tracks: factorial design with simulation-based power analysis, cross-classified Bayesian SEM bridging utterance/window/session levels, the 2x2x2 knock-out factorial isolating cascade directionality, Rosenbaum sensitivity analysis, and the ColludeBench/Colosseum collusion-detection pipeline central to H2. Dhia oversees pre-registration on OSF Registries with RFC 3161 trusted timestamps, locking hypotheses, endpoints, and stopping rules before data collection.

Research Engineer: Ahmed Dhia

Primary role: Dataset curation lead; norm erosion experiments; longitudinal analysis methodology.

Ahmed Dhia brings manufacturing engineering expertise (DFM, production scaling, ISO 13485 quality systems) plus current undergraduate training in artificial intelligence and mathematics. As Director of Manufacturing with hands-on experience in design for manufacturability, production scaling, and ISO 13485 quality systems, Dhia understands how safety-critical systems fail in practice — directly relevant to H3 norm-erosion experiments where constraint violations emerge under real-world production pressure rather than idealized lab conditions. Under this proposal, Dhia is the lead-curator-designate for the proposed 10,000+ annotated interaction-trace corpus (deliverable D6c) — to become the largest publicly available multi-agent safety dataset upon completion — and will own the MAST-compatible annotation pipeline (target kappa > 0.80), the longitudinal erosion-experiment execution under graduated pressure, and data-quality control. Dhia is completing undergraduate studies in artificial intelligence and mathematics, complementing extensive industry experience.

Two-page CVs for each PI are attached as supplementary materials.

Methodological Architecture: Five Reviewer-Replicable Instruments

Every methodological commitment in this proposal operates through five public, time-stamped, reviewer-replicable instruments: (1) **RFC 3161-stamped pre-registration** on OSF Registries for the Stage 2b protocol plus six chronologically-stamped addenda — every analytic decision is locked before data collection in a third-party verifiable timestamp chain; (2) an **independent Python/SciPy reference verifier** that re-derives the analysis pipeline from raw data and confirms numerical agreement to less than 5×10^{-3} — cross-toolchain reproduction makes the analysis falsifiable by any reader with raw data and a Python interpreter; (3) **systematic adversarial redlining** of every load-bearing claim against pre-registered falsification conditions, surfacing six required revisions prior to preprint release; (4) **pre-committed downgrade clauses** tied to data-driven decision rules that fired literally on the Stage 2b data — including retraction of an exploratory k=4 quadrimodal narrative in favor of a k=2 bimodal reading once the bootstrap procedure ran; (5) a **public Verification and Correction Log** preserving every revision so reviewers can audit the evolution of every claim. Every artifact is preserved at the public project repository (github.com/HHA-Applied-Research-Institute/colludebench-cascade).

H.H.A. is a three-person founding team operating an open-methodology research program. The five instruments above are our public review-and-verification practice — open methodology in place of a confidential advisory committee. Upon award, we engage senior external researchers as peer collaborators on specific methodological questions across the three thrust areas — Michael Wooldridge (Oxford; multi-agent systems foundations), Lewis Hammond (Cooperative AI Foundation / Oxford; cooperative-AI failure-mode taxonomy), Christian Schroeder de Witt (Oxford Witt Lab; steganographic collusion), Dylan Hadfield-Menell (MIT EECS; inverse reward design and alignment), and Sarra Alqahtani (Wake Forest; MARL safety). These are project-bounded peer consultations between professionals, scoped to specific technical questions as they arise during execution. Consultation honoraria are budgeted under BUDGET Line I.

Note on author spellings: occasional “Haidar” (for “Haedar”) and “Hassan” (for “Hass” / “Alhassan”) spelling variants in earlier RFC-3161-stamped pre-registration documents reflect pre-IRS-determination filing state. IRS-recognized legal names (1023-EZ; EIN 41-4991887) are “Haedar Hadi” and “Alhassan (Hass) Dhia.”

Preparatory Work and Risk Mitigation

This proposal funds work already underway: 13-benchmark survey, 28-paper synthesis, 347-line experimental design with simulation-based power analysis, ColludeBench-Cascade platform architecture, cascade detection dashboard prototype, eight RFC 3161 trusted-timestamps locking pre-registered analysis pipelines, cross-toolchain SciPy reference verifier returning **ALL CLAIMS REPRODUCE: True**, five-iteration named-expert adversarial Council with documented certificate state, click-to-verify claims-map at [verification/claims-map.md](#) in the public GitHub repository, and a Stage 2b preprint at github.com/HHA-Applied-Research-Institute/colludebench-cascade.

Execution risks. *Organizational capacity:* 0.5 FTE program manager; staggered hiring (core team M1-3, scale team M4-12); the documented internal review process bounds experimental scope and locks analytic claims before data-collection drift can occur. *Compute:* 36-month reserved cluster at \$2.00/GPU/hr plus burst on-demand; optional CAIS supplementary access. *Timeline:* go/no-go gates at M6, M12, M24; missed milestones trigger scope reduction to two highest-signal tracks. *Publication record:* H.H.A. is new — Stage 2b preprint preserved at the project repository with RFC 3161-stamped pre-registration, eight stamped addenda, independent SciPy reference verifier, and public Verification and Correction Log; full Stage 3 analysis plan pre-registered with RFC 3161 before submission; design builds on six validated instruments rather than novel tools, inheriting statistical properties the community already accepts. # Timeline and Milestones

The 36-month program runs five overlapping phases — infrastructure must precede experiments; individual-mode baselines must precede cross-mode interaction measurement; preliminary results must precede cross-model validation.

Phase	Months	Activity
1. Platform + Pilot	M1–6	Core team hire (M1–3); ColludeBench-Cascade architecture + middleware + topology controller; 2–8 agent pilot experiments; preliminary FAF estimates for ≥ 2 topologies
2. Core Experiments	M4–12	Class A (failure-mode elicitation, ~ 500 runs); Class B (5,000-turn longitudinal erosion); scale team hire (M4–12); preliminary FAF across all five topologies M12
3. Scale + Knock-Out	M10–24	256-agent runs; Class C governance interventions; $2 \times 2 \times 2$ knock-out factorial isolating cascade directionality; preliminary cross-failure-mode interaction coefficients M22
4. Cross-Model Validation	M18–30	Frontier-API replication (Claude, GPT-5, Gemini); collusion-detection refinement; ColludeBench-Cascade alpha M18, v1.0 M30; 10,000-trace dataset release M30
5. Dissemination	M24–36	8–12 peer-reviewed papers (NeurIPS / ICLR / ICML / ACL 2027–2029); policy briefs (EU AI Act Article 73, NIST AI RMF); community workshop; final report

Go/no-go decision milestones. *M6 — Infrastructure feasibility:* ColludeBench-Cascade operational; pilot complete; preliminary FAF for ≥ 2 topologies. Failure response: reallocate Engineer 2 budget to contract development; extend Phase 1 by 8 weeks; convene external methodology consultation for design review. *M12 — Scientific viability:* Class A produces statistically significant amplification ($p < 0.01$) for ≥ 3 topologies; Class B produces identifiable erosion dynamics for ≥ 1 model family; collusion detection achieves $> 70\%$ pre-coordination. Failure response: if fewer than two of three criteria met, reduce scope to two highest-signal tracks and reallocate from the weakest. *M24 — Generalizability:* cross-model validation confirms key findings (FAF topology dependence, erosion thresholds, or collusion detection) replicate across ≥ 2 frontier API models; preliminary cross-failure-mode interaction data available. Failure response: if findings are model-specific, pivot final 12 months toward characterizing model-specific safety profiles and publish the non-generalizability finding as an empirical contribution.

Critical-path dependencies. (1) ColludeBench-Cascade infrastructure (M1–6) gates all experimental execution. (2) Class A individual-mode baselines (M4–12) gate the knock-out cross-failure-mode design (M16–24) — interaction coefficients require individual-mode reference points. (3) Open-weight experiments (M4–24) gate frontier API validation (M18–30) — methodology validated on instrumentable models before consuming API budget where internal state is unavailable. Publication drafting, policy translation, and community engagement run in parallel and do not gate scientific progress. Staggered hiring matches personnel costs to experimental demand. # Budget and Justification

Total \$4,356,500 over 36 months: \$3,960,500 (90.9%) direct, \$396,000 indirect at exactly 9.998% of direct (\$396,000/\$3,960,500; never 10.0%), **compliant with Schmidt’s $\leq 10\%$ cap.** Personnel 69.7%, Compute (cash) 1.1%, Subawards 8.0%, Travel 3.4%, API Credits 4.0%, Platform 2.3%, remainder 2.4%. The \$4,356,500 envelope reflects the cost required to deliver the eight named outputs in §9 across a 36-month timeline: three principal investigators at 25–30% effort, three postdocs and two senior engineers, two research assistants, a half-time program manager, an $8 \times H100$ dedicated cluster with burst capacity, frontier-API validation, two subaward partners, and the open-source ColludeBench-Cascade platform.

Personnel (\$3,035,000 — 69.7%). Three principal investigators at 25–30% effort over 36 months: Lead PI Haedar Hadi at $\sim \$47K/yr$ loaded (\$140K total); Co-PI Hass Dhia at $\sim \$47K/yr$ loaded (\$140K); Research Engineer Ahmed Dhia at $\sim \$47K/yr$ loaded (\$140K). The 25–30% effort honors Schmidt’s “primary focus for lead investigator(s)” mandate at one full academic quarter per PI per year. The $\sim \$47K/yr$ loaded rate is below the NSF standard for two-month summer salary alone, indicating modest rather than inflated PI compensation. Three postdoctoral researchers (\$1.0M total, \$125K/yr loaded — 25th percentile for AI-safety postdocs at Glassdoor’s CMU benchmark): Postdoc 1 (Collusion/Game Theory, M1–36) executes ColludeBench + Colosseum + PID emergence + anti-collusion intervention evaluation directly for H2; Postdoc 2 (Norm Erosion/Longitudinal, M1–36) runs ODCV-Bench erosion curves and 5,000-turn longitudinal studies for H3; Postdoc 3 (Detection Methods, M7–30) builds the comparative evaluation framework

(paraphrase filters, governance graphs, PeerGuard, Evo-MARL). Two senior ML engineers (\$1.1M combined, \$200K/yr loaded ~20% below industry median per Levels.fyi) build the ColludeBench-Cascade platform (interception middleware, topology controller, PID pipeline, MAST trace export, 256-agent controller). Two research assistants (\$350K) and a 0.5 FTE program manager (\$165K) complete the team.

Compute (cash \$50,000 — 1.1%; capacity in-kind via Schmidt) and API Credits (\$175,000 — 4.0%). The 8×H100 80GB cluster + burst capacity for 256-agent peak periods is requested via Schmidt Sciences’ “Access to Resources” in-kind channel rather than as cash, matching the work to Schmidt’s purpose-built grantee compute infrastructure. Cash \$50K covers storage (30–40 TB NVMe + S3 cold archive), data transfer egress, and orchestration/monitoring/CI-CD infrastructure not typically bundled into in-kind GPU allocations. Equivalent commercial-cloud cash budget would be approximately \$556K over 36 months (8×H100 reserved at \$2.00/GPU/hr + ~30% utilization burst on-demand); requesting in-kind brings the cash envelope to \$4.36M, well below the named Tier-2 ceiling. Contingency: post-award CAIS supplementary-access path (no pre-submission outreach made) or cash-reactivation if both Schmidt in-kind and CAIS access are unavailable. API credits cover frontier-model **validation runs** (not factorial cells — the 5 open-weight families carry experimental rigor at zero marginal API cost) across Claude Opus 4.6 (\$5/\$25 per Mtok), Sonnet 4.6 (\$3/\$15), GPT-5.5 (\$1.75/\$14), Gemini 3 Pro (\$2/\$12), with prompt caching (90% input discount) and Batch API (50% discount) yielding \$237–\$294 per-run averaged. Stage 2b actuals confirm the envelope.

Subawards (\$350,000 — 8.0%). Two branches selected via open RFP within the first six months of award; selection criteria published as part of the institute’s open methodology. **Branch X — Game Theory (\$225K):** one postdoc + 0.5 faculty effort over 36 months at partner-institution loaded rates with overhead-waiver to Schmidt’s 10% cap; selection criteria — prior publication on multi-agent strategic interaction (Nash equilibria, mechanism design, auction theory); deliverable test — can the partner’s models predict which topologies enable collusion before experiments? **Branch Y — AI Governance and Policy Translation (\$125K):** 0.25 FTE policy researcher producing NIST AI Safety standards engagement + EU AI Act Article 73 technical-annex contributions + 2–3 policy briefs; selection criteria — documented record in AI governance / policy translation, demonstrated capacity to engage NIST or EU AI Act technical processes. Selection-after-award matches partner choice to experimental needs as they crystallize from Stage 2b → Stage 3 rather than pre-committing before the science directs the choice; Schmidt’s 10% indirect-cost cap is built into selection criteria so final partners are operationally compatible.

\$3M-Tier Fallback Scenario. If Schmidt offers a \$3M-tier rather than \$4.36M: drop Postdoc 3, Engineer 2, and RA 2; reduce PM to 0.25 FTE; eliminate burst compute; compressed frontier-API validation (400 runs across 3 frontier families); reduce travel to 2 attendees per venue; release the dataset only without the full ColludeBench-Cascade platform; produce 6–8 papers instead of 8–12; reduce external consultation engagements to ~4/year. **Direct \$2,410,000; indirect \$240,800; total \$2,650,800** (line items in BUDGET.md fallback table); \$349,200 under tier. **Honest caveat:** at \$3M-tier the program loses 256-agent scale experiments (limited to burst periods only), parallel experimental tracks (reduce 3–4 to 2), the full ColludeBench-Cascade platform (becomes dataset-only), and policy translation (governance partner cut). At \$4.36M the program is field-shaping Tier 2; at \$3M-tier it becomes excellent Tier 1. The \$1.7M differential purchases scale, platform, and policy translation.

H.H.A. Applied Research Institute is a California Public Benefit Nonprofit Corporation with 501(c)(3) status (EIN 41-4991887). Indirect costs of \$396,000 (9.09% of total / 9.998% of direct) cover accounting, financial reporting, legal compliance, audit preparation, basic administrative infrastructure, insurance, and organizational management — real operational expenses, not an artificial cap on higher actual costs. Universities typically charge 50–60% indirect rates; H.H.A.’s 10% reflects genuinely sub-10% overhead, directing approximately 1.5× the effective rate of an equivalent university proposal to direct research. # Responsible Research Plan

This research studies how AI safety failures propagate through multi-agent networks — knowledge inherently dual-use. Understanding cascade dynamics well enough to build defenses produces understanding sufficient to engineer attacks. We have built a responsible-research framework that maximizes defensive benefit while systematically reducing offensive utility.

Dual-Use Risk Categories and Mitigations

(1) Cascade propagation playbook — topology-specific FAF maps as a ranked menu of architectural vulnerability. *Mitigation:* maps published exclusively alongside validated containment protocols; raw propagation pathway diagrams under 90-day defender-first embargo. **(2) Collusion engineering via detection research** — detection signatures teach adversaries what to optimize or avoid. *Mitigation:* detection APIs released without exposing the signature library; channel-capacity estimates restricted to controlled-access appendix requiring institutional affiliation and declared defensive purpose. **(3) Agent-in-the-middle attack toolkit.** *Mitigation:* raw payloads permanently embargoed; only attack-categorization taxonomies and defense-evaluation results published; injection prompts stored in restricted repository per ACL 2025 disclosure norms. **(4) Erosion thresholds as attack scheduling.** *Mitigation:* thresh-

olds published as defensive monitoring targets without the full pressure-response curve; KPI-pressure-to-violation-rate mappings restricted to controlled access. **(5) Open-source weaponization.** *Mitigation:* platform ships with non-removable logging, rate-limited failure-injection APIs, and a research-use license; failure-injection toolkit distributed under restricted license separate from core platform.

Three-Tier Data Release

Open (immediate, ~70%): experiment configurations, aggregate metrics, statistical analyses, platform source code (Apache 2.0), topology maps paired with defenses. *90-Day Embargo (~20%):* raw experiment traces, collusion-detection threshold parameters, full cross-failure-mode interaction-coefficient models — embargo provides defenders a remediation window before detailed attack-surface information becomes public. *Controlled Access (~10%):* raw AiTM payloads (encrypted), steganographic channel-capacity by model, full pressure-to-violation curves, model-specific collusion-propensity rankings. Access requires verified institutional affiliation, stated defensive purpose, and agreement not to deploy against production systems without vendor coordination. Model-specific findings reported directly to relevant providers rather than published.

Integrity Practice (Internal Methodology)

We extend the integrity scaffold of §5.2 — RFC 3161 trusted-timestamp pre-registration, cross-toolchain SciPy reference verifier, named-expert adversarial Council review — to every claim in this program. Three additional disciplines:

Registry-consistency-check pre-commit. Every commit touching an analyzer or claim file runs an automated check asserting each load-bearing claim has a corresponding SR-M registry entry with stamped pre-registration strictly anterior to the analyzer-output commit. The check fails if a claim lacks a registry row, if the stamp is missing, or if stamp time is not strictly anterior. The same check runs in CI on the public repository.

Named-expert Council with documented certificate state. Council reviews proceed against persona panels (DataScientist + Statistician/Algorithmist, CSProfessor, Physicist, IO Domain Expert, Devil’s Advocate, Methods/Reproducibility, Consistency, Writing Quality) whose composition is documented in the Council certificate alongside iteration count, findings applied, and exit state (CLEAN, CONDITIONAL with documented user-override, or BLOCKED). Stage 2b exited CONDITIONAL after iter-5 with ~32 findings applied and unresolved findings catalogued — the certificate is auditable, not a stamp of approval.

Re-Council on transitions. When work transitions across phases (Stage 2b preprint → Schmidt proposal compression; Stage 3 dataset → cross-model paper), the Council convenes again — past CLEAN does not transfer across context. This protects against residual Lakatosian rescues that would have been visible to a fresh-context reviewer but not to one anchored in prior iterations.

Override admissibility (pre-registered). The Stage 2b preprint exited Council CONDITIONAL with a documented user-override under a venue-conditional “Schmidt-RFP-focus” filter; that override is honestly disclosed in `verification/council-certificates/review-certificate.md`. To prevent the same pattern from operating as a systemic Lakatosian protective belt across Stage 3 + Stage 4, we pre-register four override admissibility criteria binding for the funded program: (1) every override invocation lists the unresolved findings explicitly with deferral rationale (no silent overrides); (2) override-admissibility itself is audited at the next phase-transition Council, which decides whether the deferred findings remain venue-conditional or have been retroactively closed; (3) the override audit trail is published at `verification/council-certificates/` in the public repository — readers can inspect every override invocation independently; (4) any Stage 3 / Stage 4 override that touches a registered SR-M claim’s primary-source numeric target requires re-stamping the affected addendum. These criteria are themselves RFC 3161 stamped at the Stage 3 pre-registration commit and become falsifiable structural conditions on the integrity practice, not free-parameter escape valves.

Disclosure Protocol and IRB

All findings undergo three-stage pre-publication review: (1) Internal Screen (7 days) by the team’s Dual-Use Review Officer against the five risk categories — HIGH severity-and-likelihood triggers Stage 2; (2) External Reviewer Pass (30 days) — two external reviewers (one AI safety community affiliate, one cybersecurity/adversarial ML expert) recommend a release tier; (3) Affected-Party Notification (45 days) — vulnerabilities in deployed systems or frontier models trigger direct vendor notification with 90-day remediation per CERT/CC coordinated vulnerability disclosure (zero-days: 48-hour notification). The primary pipeline is automated (LLM-as-Judge, kappa = 0.77) requiring no IRB approval; ~200-trace human-annotator validation requires expedited IRB review with informed consent, content warnings, two-hour session limits with breaks, opt-out provisions, and psychological support referrals — all scenarios use synthetic data with no real patient, financial, or personal information. IRB protocol filed within 30 days of award.

Fundamental Position

Not studying multi-agent failure propagation while deployments accelerate is the greater ethical risk. Multi-agent systems are deployed at scale with documented 41–86.7% failure rates and 17.2× amplification — operating without

the empirical science of how their failure modes interact. Our framework ensures knowledge reaches defenders before adversaries, that the most dangerous specifics remain under institutional controls, and that open-source infrastructure is instrumented for safety by default. Defensive value decisively outweighs offensive knowledge produced along the way. # References

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